Teacher Answer Key — Skill Checks

Each Skill Check is out of 8 (1 mark per question, for the correct final answer). Record the mark in the box at the top of the Skill Check page and in the student's *My progress* table on the cover. A suggested pass mark for each skill is **6 / 8**.

Skill Check 1: Collecting Like Terms (8 marks, 1 per question)

Q	Question	Answer
1	$4a + 3a$	$\mathbf{7a}$
2	$9x - 5x$	$\mathbf{4x}$
3	$b + b + b$	$\mathbf{3b}$
4	$3a + 2b + 5a$	$\mathbf{8a + 2b}$
5	$6x + 4y - 2x$	$\mathbf{4x + 4y}$
6	$2x^2 + 5x^2$	$\mathbf{7x^2}$
7	$x^2 + 4x + 3x$	$\mathbf{x^2 + 7x}$
8	$3x^2 + 2x + x^2 + 5x$	$\mathbf{4x^2 + 7x}$

Marking: Accept terms in any order (e.g. 2b + 8a). The answer must be fully simplified (e.g. 3b, not b + 2b).

Skill Check 2: Multiplying Algebra (8 marks, 1 per question)

Q	Question	Answer
1	$3 \times 4a$	$\mathbf{12a}$
2	$m \times n$	$\mathbf{mn}$
3	$4x \times 3$	$\mathbf{12x}$
4	$a \times 3b$	$\mathbf{3ab}$
5	$2a \times 5b$	$\mathbf{10ab}$
6	$x \times x$	$\mathbf{x^2}$
7	$3y \times 2y$	$\mathbf{6y^2}$
8	$2x \times 4x$	$\mathbf{8x^2}$

Marking: Accept letters in any order (e.g. 3ba). Q6–8 need the square: x × x = x^2, not 2x or x × x.

Skill Check 3: Dividing Algebra (8 marks, 1 per question)

Q	Question	Answer
1	$10a \div 2$	$\mathbf{5a}$
2	$16x \div 4$	$\mathbf{4x}$
3	$xy \div y$	$\mathbf{x}$
4	$9a \div a$	$\mathbf{9}$
5	$8ab \div 4a$	$\mathbf{2b}$
6	$y^2 \div y$	$\mathbf{y}$
7	$12x^2 \div 3x$	$\mathbf{4x}$
8	$20ab \div 5b$	$\mathbf{4a}$

Marking: Accept the answer written as a simplified fraction only if it has been fully cancelled (e.g. 4x, not $12x^2/3x$).

Skill Check 4: Expanding Brackets (8 marks, 1 per question)

Q	Question	Answer
1	$4(x + 2)$	$\mathbf{4x + 8}$
2	$3(a - 5)$	$\mathbf{3a - 15}$
3	$2(3x + 1)$	$\mathbf{6x + 2}$
4	$5(2a + 3)$	$\mathbf{10a + 15}$
5	$x(x + 7)$	$\mathbf{x^2 + 7x}$
6	$y(y - 4)$	$\mathbf{y^2 - 4y}$
7	$(x + 2)(x + 3)$	$\mathbf{x^2 + 5x + 6}$
8	$(x + 1)(x + 4)$	$\mathbf{x^2 + 5x + 4}$

Marking: Accept terms in any order. Q7–8: accept the unsimplified four-term answer (e.g. $x^2 + 3x + 2x + 6$) at your discretion as partial evidence; full mark needs the like terms collected.

Skill Check 5: Factorising into Brackets (8 marks, 1 per question)

Q	Question	Answer
1	$4x + 8$	$\mathbf{4(x + 2)}$
2	$5m + 5$	$\mathbf{5(m + 1)}$
3	$3a - 9$	$\mathbf{3(a - 3)}$
4	$6y + 10$	$\mathbf{2(3y + 5)}$
5	$x^2 + 6x$	$\mathbf{x(x + 6)}$
6	$a^2 - 4a$	$\mathbf{a(a - 4)}$
7	$x^2 + 5x + 6$	$\mathbf{(x + 2)(x + 3)}$
8	$x^2 + 6x + 5$	$\mathbf{(x + 1)(x + 5)}$

Marking: Accept brackets in either order. Q4 must be fully factorised: 2(3y + 5). Students can check by expanding.

Class Record

Student	1. Collecting Like Terms / 8	2. Multiplying Algebra / 8	3. Dividing Algebra / 8	4. Expanding Brackets / 8	5. Factorising into Brackets / 8	Total / 40

Answers to the Practice Questions

Terms can be in a different order and still be right — for example $2b + 3a$ is the same as $3a + 2b$, and $(x + 3)(x + 2)$ is the same as $(x + 2)(x + 3)$.

1. Collecting Like Terms

Practice A

1. $2a$
2. $3a$
3. $5a$
4. $9x$
5. $4m$
6. $4b$
7. $5y$
8. $6x$
9. $6p$
10. $6c$
11. $2x^2$
12. $7x^2$

Practice B

1. $6a + 3b$
2. $6x + 2y$
3. $4a + 6b$
4. $6x + 3$
5. $5m + 3n$
6. $2a + 2b$
7. $4x + 5$
8. $5x + 2y$
9. $6ab$
10. $x^2 + 5x$
11. $3x^2 + x$
12. $4y^2 + 6y$

2. Multiplying Algebra

Practice A

1. $4a$
2. $5x$
3. ab
4. xy
5. $6a$
6. $12x$
7. $15m$
8. $12x$
9. a^2
10. y^2
11. $2ab$
12. pqr

Practice B

1. $6ab$
2. $8xy$
3. $10ab$
4. $6ab$
5. $3a^2$
6. $10x^2$
7. $12m^2$
8. $6x^2$
9. $6ab$
10. x^3
11. x^3
12. $6x^3$

3. Dividing Algebra

Practice A

1. $4a$
2. $4x$
3. $2b$
4. $5y$
5. $3x$
6. $3p$
7. $2x$
8. m
9. a
10. y
11. 5
12. 7

Practice B

1. $3b$
2. $3x$
3. $2a$
4. $4y$
5. $2ab$
6. x
7. a
8. $4x$
9. $3x$
10. $3a$
11. $5x$
12. $3x$

4. Expanding Brackets

Practice A

1. $2x + 6$
2. $3a + 12$
3. $5x + 5$
4. $6x + 12$
5. $4y - 8$
6. $2m - 10$
7. $6x + 3$
8. $8a + 6$
9. $10y - 5$
10. $x^2 + 2x$
11. $a^2 + 5a$
12. $x^2 - 3x$

Practice B

1. $x^2 + 3x + 2$
2. $x^2 + 7x + 12$
3. $x^2 + 7x + 10$
4. $x^2 + 7x + 6$
5. $x^2 + 6x + 8$
6. $x^2 + 6x + 5$
7. $x^2 + 6x + 9$
8. $x^2 + 9x + 20$

5. Factorising into Brackets

Practice A

1. $3(x + 2)$
2. $2(a + 4)$
3. $5(x + 3)$
4. $7(m + 1)$
5. $4(y - 3)$
6. $2(x - 5)$
7. $3(2a + 3)$
8. $5(2x + 3)$
9. $x(x + 3)$
10. $a(a + 5)$
11. $x(x - 2)$
12. $y(y + 1)$

Practice B

1. $(x + 1)(x + 2)$
2. $(x + 1)(x + 3)$
3. $(x + 2)(x + 4)$
4. $(x + 1)(x + 4)$
5. $(x + 2)(x + 5)$
6. $(x + 3)(x + 4)$
7. $(x + 3)(x + 5)$
8. $(x + 3)(x + 3)$

★ Mixed Review

1. $7a$
2. $6x$
3. $4y$
4. $2x + 10$
5. $3(x + 4)$
6. $6x + 3y$
7. $4ab$
8. $5a$
9. $x^2 + x$
10. $x(x + 7)$
11. $5m$
12. $5x^2$
13. $6x$
14. $8a - 4$
15. $5(y - 4)$
16. $3x^2 + 3x$
17. $8a^2$
18. $2y$
19. $x^2 + 4x + 3$
20. $(x + 2)(x + 2)$